Vinay Kudari

Applied research on agentic systems, evaluation, and self-improvement

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Experience

- **Google** — San Francisco, CA
  Software Engineer, Agentic AI — *July 2026 - Present*
  - Conduct applied research on agent behavior modeling and trajectory-level analysis; characterize recurring failure modes and inefficiencies to inform hypotheses for harness improvements and model post-training
  - Design and build evaluation environments and task suites for a custom agent harness to measure task success and trajectory quality, compare harness changes, and detect behavioral regressions
  - Develop a recursive self-improvement (RSI) loop that translates agent execution feedback into proposed harness changes and evaluates candidate improvements for Google Cloud workflows
- **Apple** — Cupertino, CA
  Software Engineer | ***fine-tuning, inference, agents*** — *July 2024 - June 2026*
  - Engineered an enterprise NL-to-SQL pipeline over 10K+ warehouse tables; fine-tuned a 7B Qwen2.5 planner with FSDP on 100K+ samples, using retrieval and negative rejection to ground JSON query plans
  - Delivered a policy-aware invoice intelligence system combining receipt parsing, Snowflake policy retrieval, and evidence-backed validation to automate processing of 5K+ invoices/day
  - Designed a deep-research finance analyst agent that retrieves from databases and flat files to produce dashboards, slide decks, and cited financial insights from natural-language requests
- **Gap Inc** — San Francisco, CA
  Senior Software Engineer | ***ranking, retrieval, e-commerce*** — *July 2023 - June 2024*
  - Spearheaded development of a real-time personalization engine using user access patterns to tailor the e-commerce experience
  - Engineered a low-latency LLM-based retrieval system to extract information from a multi-source document corpus
- **Amazon Web Services** — Palo Alto, CA
  Software Development Engineer | ***data platforms, distributed systems*** — *Dec 2022 - June 2023*
  - Designed and optimized complex SQL queries in Athena resulting in a 40% reduction in query execution times
  - Architected near real time data processing system using AWS CDK, leveraging PySpark for ETL transforms on multi-terabyte healthcare data sets
- **Brave Orbit (early stage startup)** — Cape Town, South Africa
  Founding Engineer | ***Django, Flutter*** — *Jan 2020 - Aug 2021*
  - Built a cross-platform healthcare app and backend services for medication reminders and recurring clinical surveys, including asynchronous notifications, scheduling, and response collection

Projects

- **Pixie**: Built in 6 hours at the Nano Banana Hackathon; enables selective edits on live website screenshots or mockups using natural language or voice, with multi-model generation and real-time canvas updates
- **Friend (@mydawg_bot on telegram)**: Building a multi-agent AI companion with human-like memory simulation, capable of personalized conversations, context-aware recall, and tool use
- **Stocklerts**: Built a self-evolving system that analyzes real-time news to recommend stocks for day trading; adapts prompts weekly based on ground-truth top performers to refine future predictions
- **Open-source contributions**: Contributor to HuggingFace Datasets official repository

Relevant skills

Languages: Python, C++, SQL
ML & Deep Learning: PyTorch, JAX, Hugging Face, PySpark
Agentic Systems: Agent evaluation, trajectory analysis, LangGraph, vLLM, RAG
Infra & Backend: Kubernetes, FastAPI, Redis, Snowflake, GCP, AWS

Education

- **University at Buffalo** — Buffalo, NY
  Master of Science in Computer Science (AI/ML Specialization)
  Research: *Finding inconsistency between text and related images, advised by Prof. David Doermann*
- **National Institute of Technology, Durgapur** — West Bengal, India
  Bachelor of Technology in Computer Science and Engineering